

MedEasy Pharmacy OS

Complete QA Testing Guide, UAT Checklist & Operations Manual

Frontend URL:	https://beta-2-seven.vercel.app
Backend API:	https://beta2-h6y2.onrender.com (Node.js + PostgreSQL)
Admin Credentials:	Username: admin Password: admin123
Alternate Admin:	Username: ninaad_nk Password: password123
Registered Mobile:	8380036778 (For password reset verification)

1. Authentication & Security Testing (Page: /login)

Eye Toggle Button (Show/Hide Password)	1. Open /login 2. Type password into the password field 3. Click the Eye Icon on the right side	Password switches from masked dots (•••) to visible plain text. Clicking again hides it back.
Admin Login	Enter username: admin , password: admin123 , click 'Login to Counter'	Authenticates instantly, receives JWT Bearer token, redirects directly to /dashboard .
Forgot / Reset Password Flow	1. Click 'Forgot password?' link on login page 2. Enter username: admin 3. Enter mobile: 8380036778 4. Enter new password & click eye icon	Verifies registered phone against database, resets password with success alert, auto-fills login form.
Role-Based Access Control (RBAC)	Log in with a STAFF account and try opening /settings or /purchases	Protected route guard automatically redirects unauthorized roles back to /dashboard .

2. POS Billing Counter Testing (Page: /billing)

Feature / Action	Exact Steps to Test	Expected Behavior
1,000+ Master Drug Live Search	In billing search bar, type: Dolo , Pan , or Azithral	Autocomplete dropdown appears instantly showing both in-stock batches and master catalog medicines.
1-Click Billing from Master Catalog	Click '+ 1-Click Bill' on any un-inwarded master medicine (e.g. Pan 40)	Auto-populates cart with current selling price, MRP, and 12% GST rate without requiring manual entry.
Mobile Customer Selector Bar	Open /billing on mobile screen (< 640px) or shrink browser window	Clean card layout: customer label & '+ New Customer' button on top, full-width select dropdown, zero overflow.
Schedule H / Doctor Rx Details	Enter Prescribing Doctor: <i>Dr. Kulkarni MBBS</i> and Patient Name: <i>Suresh Patil</i>	Saves doctor name with the invoice and satisfies FDA Schedule H statutory requirement.
Thermal 80mm POS Print	Add items to cart and click 'Print (F9)'	Opens 80mm formatted POS receipt with store header, DL number, GSTIN, line items, and signature line.
WhatsApp Digital Bill	Click 'WhatsApp Bill', confirm customer mobile number	Opens WhatsApp Web / Mobile with a pre-formatted tax invoice summary ready to send.

3. Sales Invoices History, Reprints & Returns (Page: /sales)

Feature / Action	Exact Steps to Test	Expected Behavior
Browse Past Invoices	Navigate to /sales via sidebar link 'Sales History'	Displays full table of historical sales fetched from PostgreSQL database.
Search & Payment Filters	Filter by: Cash , UPI , or Card ; or search by invoice #	Instantly filters table rows and displays payment breakdown metrics.
Thermal Reprint Copy	Click the Printer Icon on any past invoice row	Opens print dialog with exact 80mm receipt marked with '*** REPRINT COPY ***'.
Resend WhatsApp Bill	Click the Icon on any past invoice row	Opens WhatsApp with that specific past invoice breakdown.
Bill Cancellation & Stock Reversal	Click the Rotate Icon (■) on an invoice, confirm cancellation	Atomic rollback: marks bill CANCELLED, restores item quantities back into batch inventory, deducts customer spend.

4. Inventory & Batch Vault Testing (Page: /stock)

Feature / Action	Exact Steps to Test	Expected Behavior
Stock Filter Tabs	Click tabs: All Stock , Low Stock , and Expiring Soon	Filters inventory in real-time by current stock quantity and expiry date (< 90 days).
Batch Management	Click on any medicine name (e.g. Dolo 650) to open details page	View all batches sorted by First-Expiry-First-Out (FEFO), add new batch, or adjust stock.
Physical Stock Adjustment	Click 'Adjust Stock' on a batch, enter new count and reason	Updates batch stock and appends a permanent audit adjustment log.
CSV Stock Export	Click the 'CSV' button in top toolbar	Downloads a complete spreadsheet of all store medicines, batches, and stock counts.

5. Distributor Purchases & Suppliers (Pages: /purchases & /suppliers)

Feature / Action	Exact Steps to Test	Expected Behavior
Add Distributor / Supplier	On /suppliers, click '+ Add Supplier', enter name, GSTIN, mobile	Saves distributor to PostgreSQL database and makes them selectable in purchases.
Inward Purchase Invoice	On /purchases, select distributor, enter invoice number, add medicine item with batch, expiry & cost	Saves purchase invoice and automatically increments warehouse stock in the inventory vault.
Purchase Invoices History	Click the 'Purchase Invoices History' tab on /purchases	Lists all received distributor purchase bills with supplier name and total amount.

6. Customer Management & Chronic Refill Alerts (Page: /customers)

Feature / Action	Exact Steps to Test	Expected Behavior
Customer Directory & Spend	Browse customer list on /customers	Tracks customer mobile, lifetime purchase value (■), and total number of bills.
Chronic Refill Reminders Tab	Click tab '■ Chronic Refill Reminders'	Filters clients who have active prescription bills and are due for their monthly refill.

1-Click WhatsApp Refill Alert	Click the '■ Refill Alert' button on a customer row	Opens WhatsApp with personalized reminder inviting them to reserve their medicines.
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7. Financial Reports, GST Audit & Schedule H Register (Page: /reports)

Feature / Action	Exact Steps to Test	Expected Behavior
Financial Overview	Open /reports	Displays real Sales Turnover, Purchase Spend, Invoices count, and monthly sales bar chart.
GST Tax & GSTR-1 CSV Export	Click tab 'GST Tax & GSTR-1 Filing', then click 'Download GSTR-1 CSV'	Calculates Taxable Turnover, CGST (6%), SGST (6%), Input Tax Credit (ITC), and exports CSV for accountant/CA.
Schedule H / Rx Inspection Register	Click tab 'Schedule H / Rx Register', search doctor name, click 'Export CSV'	Gathers prescription sales with patient, doctor, batch, and qty for FDA Drug Inspector compliance audit.

8. Pharmacy Store Profile & Settings (Page: /settings)

Feature / Action	Exact Steps to Test	Expected Behavior
Store License Persistence	Edit Store Name, Drug License (DL), GSTIN, Phone, Address; click 'Save Store Profile & Licenses'	Saves permanently to PostgreSQL. Refreshes preserve values, and thermal receipts automatically print the updated details.
Staff User Registration	Click '+ Add Staff User', enter name, username, password, select role (PHARMACIST/STAFF)	Creates active user in database; new staff member can immediately log in from the login page.
Language Toggle (English / Marathi)	Click the language toggle in top header or settings (EN / MR)	Entire UI instantly translates to Marathi (■■■■■) or English.

Quality Assurance Verification Sign-off:

All 30 automated integration tests passed (30/30). All frontend views build with 0 TypeScript/Vite errors. The application is completely functional and ready for daily medical counter operations.